

Validity Anomaly Types

Data Quality Platform — Complete reference of all anomaly signals

Generated May 13, 2026

Score legend

■ ≥ 0.5 High

■ ≥ 0.3 Medium

■ < 0.3 Low

Scores combine as: $\text{row_score} = 1 - \prod(1 - \text{score}_i)$ for scores ≥ 0.15

■ Null

Anomaly	Score	Meaning
null_not_allowed	1.0	Null in a non-nullable column
rare_null	0.15	Null where null rate is historically $< 1\%$

■ Numeric

Anomaly	Score	Meaning
extreme_outlier	0.7	Z-score > 12 , outside p01–p99
strong_outlier	0.35	Z-score > 6 , outside p01–p99
parse_fail	0.3	Value cannot be parsed as a number
moderate_outlier	0.1	Z-score > 3 , or extreme/strong but within p01–p99 (likely sentinel code)
outside_p01_p99	0.1	No Z-score signal but outside historical percentile range

■ Categorical

Anomaly	Score	Meaning
unseen_value	0.35	Value never seen before, low-cardinality column
rare_value	0.25	Value seen but frequency $< 0.1\%$
unseen_value_high_cardinality	0.1	Value never seen before, high-cardinality column
low_frequency	0.1	Value seen but frequency between 0.1% – 1%

■ Text / Structured

Anomaly	Score	Meaning
shape_violation	0.5	Column has a dominant format and this value breaks it
length_anomaly	0.25	Length deviates more than 6 standard deviations from mean
unseen_shape	0.1	Shape pattern never seen before, no dominant format
rare_shape	0.1	Shape seen but frequency < 1%
length_deviation	0.1	Length deviates more than 3 standard deviations from mean
unseen_shape_soft	0.05	Free-text column, shape never seen before (very lenient)

■ Datetime

Anomaly	Score	Meaning
unexpected_future	0.5	Future date when historically < 1% of values are future
parse_fail	0.3	Value cannot be parsed as a date
far_before_historical_min	0.35	More than 50% of the date range before the earliest seen date
far_after_historical_max	0.2	More than 50% of the date range after the latest seen date
rare_future	0.2	Future date when historically 1%–5% of values are future
before_historical_min	0.1	Before earliest seen date, within 50% of the date range
after_historical_max	0.1	After latest seen date, within 50% of the date range

■ Boolean

Anomaly	Score	Meaning
unexpected_false	0.4	False in a column that is ≥ 99% True
unexpected_true	0.4	True in a column that is ≤ 1% True
rare_false	0.2	False in a column that is ≥ 95% True
rare_true	0.2	True in a column that is ≤ 5% True

Row score threshold (ROW_SCORE_THRESHOLD) controls which rows are flagged. Scores below MIN_CONTRIBUTION (0.15) are recorded in details but do not accumulate into the row score.